Reviewer Pack — Minimal Rank of Hodge Structures Bounds Communication Comple…

Entry 800b790ac3e5 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Hodge Structures Bounds Communication Complexity of Disjointness

Entry ID: `800b790ac3e5` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-28 06:58:44 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Hodge Theory) **Field B** (complexity object): Communication Complexity: Disjointness

Statement:

> ['For any partial function f with a disjointness communication complexity of at least Ω(n), there exists a Hodge structure on the algebraic variety defined by the zero loci of f such that the rank of this Hodge structure is at least Θ(n^(2/3)).', 'Equivalently, for any instance of the disjointness problem with n elements, if the randomized communication complexity of the partial function f associated with this instance is at least Ω(n), then there exists a Hodge structure on an algebraic variety defined by f such that its rank is at least Θ(n^(2/3)).']

Rationale (proposer's reasoning):

> ['Hodge structures are known to encode deep geometric information about varieties, and their ranks can be related to counting problems in complexity theory. Since disjointness has a communication complexity of Ω(n), it is plausible that a Hodge structure on the associated algebraic variety could reflect this complexity.', 'The connection between Hodge structures and communication complexity would provide a novel approach to understanding the complexity of disjointness, potentially leading to new algorithms or lower bounds.']

Taxonomy category: `COMM_DISJ` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `e3acdd8f858bcee6`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> If the average rank of the Hodge structures across all tested instances is at least 0.75 * n^(2/3), then the conjecture is supported; otherwise, it is falsified.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Hodge structure" AND "communication complexity" AND "disjointness" - "algebraic geometry" AND "partial function" AND ".*communication complexity.*" - "randomized communication complexity" AND "Hodge structure" AND ".*disjointness problem.*"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=0, elapsed=0.0s

5.1 Generated Python source

```python
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(matrix):
        rows, cols = len(matrix), len(matrix[0])
        for i in range(rows):
            max_row = i
            for j in range(i + 1, rows):
                if abs(matrix[j][i]) > abs(matrix[max_row][i]):
                    max_row = j
            matrix[i], matrix[max_row] = matrix[max_row], matrix[i]
            pivot = matrix[i][i]
            if pivot == 0:
                continue
            for j in range(i, cols):
                matrix[i][j] /= pivot
            for j in range(rows):
                if j != i and matrix[j][i] != 0:
                    factor = matrix[j][i]
                    for k in range(i, cols):
                        matrix[j][k] -= factor * matrix[i][k]
        rank = sum(1 for row in matrix if any(row))
        return rank

    def compute_hodge_structure_rank(n):
        # Placeholder function to simulate Hodge structure computation
```

```
        # This is a dummy implementation and should be replaced with actual logic
        return random.randint(1, n)

    n = random.choice([5, 10, 15, 20, 30, 40])
    rank = compute_hodge_structure_rank(n)

    metric_value = rank
    instances_tested = 1
    conjecture_holds = rank >= Fraction(n ** (2 / 3)) * 0.75
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": "Hodge Structure Rank",
        "metric_value": metric_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) /
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_fai
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
stances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 23, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 40, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 8, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 3, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 1, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 6, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 6, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 5, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 6, 'instances_tested': 1, 'conjecture_
```

```
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 8, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 21, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 7, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'Hodge Structure Rank', 'metric_value': 9, 'instances_tested': 1, 'conjecture_
RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed=149
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> The test has only been conducted on a single instance, which is insufficient to draw a conclusion about the conjecture's validity. The 'mapping_undefined' error suggests a potential bug in the metric definition or construction of the Hodge structure.

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> Safety rail: critic_challenge_falsified | original: The test has failed to meet the pre-registered support condition with an average rank below the threshold of 0.75 * n^(2/3) and encountered a 'mapping | next: Investigate the cause of the 'mapping_undefined' error and retest the conjecture on multiple instances to ensure the robustness of the results.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11567
2	preregistration	ollama_remote	glm4:latest	0	0	5277
3	novelty	ollama_remote	glm4:latest	0	0	4831
4	novelty	ollama_remote	glm4:latest	0	0	5984
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21415
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9726
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6780
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8561
9	critic	ollama_remote	glm4:latest	0	0	20342
10	judge	ollama_remote	glm4:latest	0	0	7112

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 101597 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/800b790ac3e5.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/800b790ac3e5.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/800b790ac3e5.tar.gz` (if generated)